

Activity 2 — Web Scraping with Python

What you'll learn

Web scraping means pulling data straight off a web page with code. You'll scrape the **CAR Region Open Data Portal** — a live web page with three tables (weather stations, schools, and tourism sites) — do a **first pass of data cleaning** (handle missing values and duplicates), and build an analysis dashboard.

The page we scrape: https://datasciencebagueio-production.up.railway.app/datasets/car_data

Honest note: this portal holds **simulated sample data** made for this workshop. The numbers are realistic but invented — not official government figures. What's real is the **skill**.

Before you start

- A **Google account** to use Google Colab.
- An **internet connection** (we pull the page live — no file to upload).

Run each cell with **Shift + Enter**, top to bottom.

Step 1 — Install the scraping tools

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```
!pip install beautifulsoup4 requests
```

You should see: install messages or "Requirement already satisfied".

Step 2 — Load the libraries

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```
import requests
from bs4 import BeautifulSoup
import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import display

print("Libraries loaded.")
```

You should see: Libraries loaded.

Step 3 — Download the web page

`requests.get` fetches the raw page from the portal. A status of **200** means success.

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```
url = 'https://datasciencebagueio-production.up.railway.app/datasets/car_data'

try:
    response = requests.get(url, timeout=15)
    print(f"Status code: {response.status_code}")
    if response.status_code == 200:
        print("Page downloaded successfully.")
    else:
        print("Got a response, but not 200. The site may be busy.")
except Exception as e:
    print(f"Could not reach the site: {e}")
    print("Check your internet connection and try again.")
```

You should see: Status code: 200 and "Page downloaded successfully."

Step 4 — Parse the HTML and look at the page

BeautifulSoup turns the raw HTML into something we can search. We read the page heading, the notice, and count how many tables are on the page.

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```

soup = BeautifulSoup(response.text, 'html.parser')

print("Page heading:", soup.find('h1').text)

notice = soup.find('p')
if notice:
    print("\nNotice on the page:")
    print(" ", notice.text.strip())

tables_on_page = soup.find_all('table')
print(f"\nThis page has {len(tables_on_page)} tables.")

print("\nSection titles found:")
for h2 in soup.find_all('h2'):
    print(" -", h2.text)

```

You should see: the page heading, the "simulated data" notice, "This page has 3 tables", and the three section titles.

Step 5 — Scrape all three tables at once

`pd.read_html` reads every table on the page into a list. This page has three.

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```

tables = pd.read_html(response.text)
print(f"Scraped {len(tables)} tables from the page.\n")

weather_df = tables[0]
schools_df = tables[1]
tourism_df = tables[2]

print(f"Weather stations: {len(weather_df)} rows")
print(f"Schools: {len(schools_df)} rows")
print(f"Tourism sites: {len(tourism_df)} rows")

```

You should see: Weather **47**, Schools **51**, Tourism **40** rows. (The weather and schools counts look a little high — there's a duplicate row hiding in each. We'll catch it in Step 7.)

Step 6 — Look at what we scraped

Always eyeball your data before analyzing it.

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```
print("WEATHER STATIONS (first 5):")
display(weather_df.head())
print("SCHOOLS (first 5):")
display(schools_df.head())
print("TOURISM SITES (first 5):")
display(tourism_df.head())
```

You should see: the first 5 rows of each table.

Step 7 — Check the data quality

Real scraped data is rarely perfect. Before analyzing, check each table for **missing values** (empty cells) and **duplicate rows**.

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```
for name, df in [('WEATHER', weather_df), ('SCHOOLS', schools_df), ('TOURISM',
tourism_df)]:
    print(f"{name}: {df.isnull().sum().sum()} missing cells, {df.duplicated().sum()}
duplicate rows")

print("\nMissing cells per column (weather):")
print(weather_df.isnull().sum()[weather_df.isnull().sum() > 0])

print("\nMissing cells per column (schools):")
print(schools_df.isnull().sum()[schools_df.isnull().sum() > 0])
```

You should see: Weather and Schools each report some missing cells and **1 duplicate row**; Tourism is clean (0 and 0).

Step 8 — A first pass at cleaning

The tourism table is already clean, but **weather** and **schools** have a few empty cells and a duplicate row. We do a light, sensible clean-up:

- **Remove duplicate rows** (the exact same record appearing twice).
- **Fill missing numbers** with the column **average**.
- **Fill missing text** with the **most common value** (the mode).

This is just a **first pass**. Full cleaning (outliers, data types, validation) is a bigger topic for later.

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```
# 1) Remove duplicate rows
weather_df = weather_df.drop_duplicates().copy()
schools_df = schools_df.drop_duplicates().copy()

# 2) Fill missing NUMBERS with the column average
for col in ['Humidity_pct', 'Rainfall_mm']:
    weather_df[col] = weather_df[col].fillna(round(weather_df[col].mean()))
schools_df['Teachers'] =
schools_df['Teachers'].fillna(round(schools_df['Teachers'].mean()))
schools_df['Year_Established'] =
schools_df['Year_Established'].fillna(schools_df['Year_Established'].median())

# 3) Fill missing TEXT with the most common value (mode)
weather_df['Condition'] = weather_df['Condition'].fillna(weather_df['Condition'].mode()[0])
schools_df['Type'] = schools_df['Type'].fillna(schools_df['Type'].mode()[0])

print("After the first cleaning pass:")
for name, df in [('WEATHER', weather_df), ('SCHOOLS', schools_df), ('TOURISM', tourism_df)]:
    print(f" {name}: {len(df)} rows, {df.isnull().sum().sum()} missing cells, {df.duplicated().sum()} duplicates")

print("\nNote: this is a first pass only. Full cleaning comes later.")
```

You should see: after cleaning, Weather back to **46** rows, Schools **50** rows, and **0 missing cells, 0 duplicates** everywhere.

Step 9 — Quick summary numbers

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```
print("WEATHER")
print(f"  Average temperature: {weather_df['Temperature_C'].mean():.1f} C")
print(f"  Average rainfall: {weather_df['Rainfall_mm'].mean():.1f} mm")

print("\nSCHOOLS")
print(f"  Total enrollment: {schools_df['Enrollment'].sum():,} students")
print(f"  Average class ratio: {schools_df['Student_Teacher_Ratio'].mean():.1f} students per teacher")

print("\nTOURISM")
print(f"  Total annual visitors: {tourism_df['Annual_Visitors'].sum():,}")
print(f"  Average rating: {tourism_df['Rating'].mean():.2f} out of 5")
```

You should see: headline figures for each table (e.g. avg temperature ~24 C).

Step 10 — Correlation between two variables

A **correlation** tells us if two number columns move together. It ranges from **-1 to +1**: near **+1** they rise together, near **-1** one rises as the other falls, near **0** there's no clear relationship. We check **elevation** vs **temperature** in the weather data.

In real life we'd expect higher places to be cooler (a negative correlation). Our data is randomly simulated, so don't over-read the exact number — the point is learning *how* to measure a relationship.

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```

x = 'Elevation_m'
y = 'Temperature_C'

r = weather_df[x].corr(weather_df[y])
print(f"Correlation between {x} and {y}: {r:.3f}")

if r > 0.5:
    print("That's a fairly strong POSITIVE relationship.")
elif r < -0.5:
    print("That's a fairly strong NEGATIVE relationship.")
else:
    print("That's a weak relationship (close to zero).")

plt.figure(figsize=(8, 5))
plt.scatter(weather_df[x], weather_df[y], alpha=0.6, color='purple')
plt.title(f'{x} vs {y} (correlation = {r:.2f})', fontweight='bold')
plt.xlabel('Elevation (m)')
plt.ylabel('Temperature (C)')
plt.grid(alpha=0.3)
plt.show()

```

You should see: a correlation value (somewhere around -0.4 on this sample) and a scatter plot of the two variables.

Step 11 — The analysis dashboard

Six charts in one figure, pulling from all three scraped tables.

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```

fig, axes = plt.subplots(2, 3, figsize=(18, 10))
fig.suptitle('CAR Region Data Dashboard (Simulated Sample Data)', fontsize=15,
fontweight='bold')

# 1. Temperature by province
weather_df.boxplot(column='Temperature_C', by='Province', ax=axes[0, 0])
axes[0, 0].set_title('Temperature by Province')
axes[0, 0].set_xlabel('')
axes[0, 0].tick_params(axis='x', rotation=45)

# 2. Students by school type
by_type = schools_df.groupby('Type')['Enrollment'].sum()
axes[0, 1].pie(by_type.values, labels=by_type.index, autopct='%1.0f%%', startangle=90)
axes[0, 1].set_title('Students by School Type')

# 3. Rainfall vs elevation
axes[0, 2].scatter(weather_df['Elevation_m'], weather_df['Rainfall_mm'], alpha=0.6,
color='teal')
axes[0, 2].set_title('Rainfall vs Elevation')
axes[0, 2].set_xlabel('Elevation (m)')
axes[0, 2].set_ylabel('Rainfall (mm)')

# 4. Tourism ratings
axes[1, 0].hist(tourism_df['Rating'], bins=12, color='gold', edgecolor='black')
axes[1, 0].axvline(tourism_df['Rating'].mean(), color='red', linestyle='--',
label=f"Mean {tourism_df['Rating'].mean():.2f}")
axes[1, 0].set_title('Tourism Site Ratings')
axes[1, 0].set_xlabel('Rating')
axes[1, 0].legend()

# 5. Visitors by tourism type
by_t = tourism_df.groupby('Type')['Annual_Visitors'].sum().sort_values()
axes[1, 1].barh(by_t.index, by_t.values, color='coral')
axes[1, 1].set_title('Annual Visitors by Type')
axes[1, 1].set_xlabel('Visitors')

# 6. Weather conditions
cond = weather_df['Condition'].value_counts()
axes[1, 2].bar(range(len(cond)), cond.values, color='lightblue')
axes[1, 2].set_xticks(range(len(cond)))
axes[1, 2].set_xticklabels(cond.index, rotation=45, ha='right')
axes[1, 2].set_title('Weather Conditions')

```



```
axes[1, 2].set_ylabel('Stations')

plt.tight_layout()
plt.show()
print("Dashboard built from the scraped CAR data.")
```

You should see: a 6-chart dashboard (boxplot, pie, scatter, histogram, bar, bar).

Tips & common errors

- Run cells **top to bottom** with Shift + Enter.
- **Status not 200, or a connection error?** Check your internet and re-run Step 3. The portal may be waking up — try once more.
- **KeyError on a column?** Column names are spelling- and case-sensitive (e.g. Temperature_C, Annual_Visitors). Use them exactly.
- Always check a site's **terms** before scraping it, and always be clear whether data is **real or simulated**.

What's next

In **Activity 3** you'll take this scraped tourism data and run a full **exploratory data analysis**.